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Robustness of Deep Learning models in electrocardiogram noise detection and classification

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ABSTRACT

Background and Objective : Automatic electrocardiogram (ECG) signal analysis for heart disease detection has gained significant attention due to busy lifestyles. However, ECG signals are susceptible to noise, which adversely affects the performance of ECG signal analysers. Traditional blind filtering methods use predefined noise frequency and filter order, but they alter ECG biomarkers. Several Deep Learning-based ECG noise detection and classification methods exist, but no study compares recurrent neural network (RNN) and convolutional neural network (CNN) architectures and their complexity.

Methods: This paper introduces a knowledge-based ECG filtering system using Deep Learning to classify ECG noise types and compare popular computer vision model architectures in a practical Internet of Medical Things (IoMT) framework. Experimental results demonstrate that the CNN-based ECG noise classifier outperforms the RNN-based model in terms of performance and training time.

Results: The study shows that AlexNet, visual geometry group (VGG), and residual network (ResNet) achieved over 70% accuracy, specificity, sensitivity, and F1 score across six datasets. VGG and ResNet performances were comparable, but VGG was more complex than ResNet, with only a 4.57% less F1 score.

Conclusions: This paper introduces a Deep Learning (DL) based ECG noise classifier for a knowledge-driven ECG filtering system, offering selective filtering to reduce signal distortion. Evaluation of various CNN and RNN-based models reveals VGG and Resnet outperform. Further, the VGG model is superior in terms of performance. But Resnet performs comparably to VGG with less model complexity.

1. Introduction

Electrocardiogram (ECG) is the most commonly used bio-marker for identifying heart abnormalities and a potential marker that can be used for the early detection of heart disease. However, the ECG collected using wearable sensors in daily living conditions are prone to noise due to changes in posture, the intensity of activities and other movements [1,2]. The use of noisy ECG in automated decision-making systems can result in poor clinical decision i.e., an increase in false alarm rate or a decrease in sensitivity, which make these systems impractical to use [3,4]. Therefore, assessing ECG signal quality is the key for developing any automated prediction, diagnostic or monitoring system.

There are several ways to quantify the ECG signal quality. For example, threshold-based ECG noise detection, called statistical signal quality indices ($SSQIs$), are proposed to detect the presence of noise in ECG signal [5,6]. However, these approaches use empirical thresholds to determine the presence of noise, which is difficult to generalise across different datasets [6,7].

To address this limitation, classical Machine Learning (ML), such as support vector machine (SVM), k-nearest neighbours (KNN), and adaptive boosting (AdaBoost), was used to detect ECG noise using statistical features where the model learnt the pattern instead of setting a fixed threshold values [8]. However, performances of these classical ML models largely depend on the quality of feature extraction and selection process [9]. In addition, feature extraction sometimes becomes computationally expensive, especially for resource-constrained wearable devices with limited processing power and energy supply [10].

In this context, Deep Learning (DL) provides an alternative approach to address the limitations of handcrafted feature extraction limitation by learning important features directly from raw ECG signals [11–14]. Similar to the classical ML model, DL also removes the dependency on a threshold value. By leveraging DL, the need for manual feature engineering is minimised, making it more suitable for efficiently

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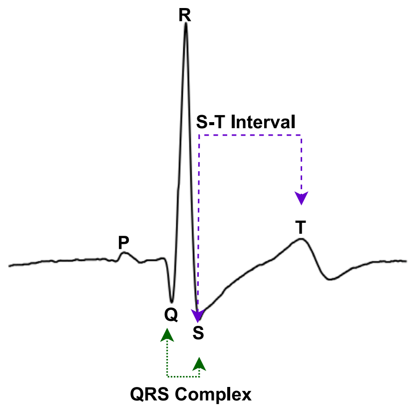

Fig. 1. Visualisation of ECG signal with clinical biomarkers, such as QRS complex and S-T interval.

processing ECG data. However, one problem with noise detection in DL-based ECG analysis is that DL models struggle to generalise to unseen noise patterns or variations, potentially leading to misclassification or reduced accuracy in noisy conditions. Ensuring robustness and adaptability to diverse noise sources remains challenging in DL-based ECG noise detection systems. Another limitation is that there is no further information about how this DL-based ECG noise detection system can used for smart filtering systems instead of blind filters.

ECG signal filtering is one of the techniques to remove noise from the signals. However, blind filtering distorts vital clinical ECG biomarkers, such as QRS complex and ST interval (as shown in Fig. 1). To tackle critical clinical ECG biomarker distortion, it is essential to use knowledge-based filtering through ECG noise detection and classification. There are three methods for ECG noise detection and classification: (i) threshold-based, (ii) ML-based, and (iii) DL-based ECG noise detection classification, which aim to mitigate biomarker distortion. However, a majority of these existing methods have relied on simulated datasets [15,16], private datasets [17], or a limited selection of public datasets for validation [8]. Additionally, the existing studies often employed their proposed DL architectures, lacking broader validation and widespread applicability (a detailed discussion in Section 1.2). The aforementioned limitations raise concerns about these DL models' robustness and generalisation capabilities. There is a need to focus on exploring the robustness of DL models in ECG noise detection and classification. As a result, we motivate to compare various DL models for ECG noise detection and classification without introducing a new model. We aim to highlight that a model's performance is not the only consideration; factors like parameters and model size are also crucial. These aspects aid in creating efficient DL-based ECG noise detection and classification systems, for example, an Internet of Medical Things (IoMT) scenario (as discussed in Section 1.1).

That said, this study aims to assess the performance, complexity, and adaptability of computer vision architectures in 1d ECG noise classification across various datasets. Concurrently, we aim to increase the efficacy of DL-based ECG noise detection and classification to develop robust knowledge-based ECG filtering methods. This knowledge-based ECG filtering will improve the accuracy and stability of disease classification, notably in arrhythmia and stress detection under real-life conditions. Moreover, combining these methods within an IoMT framework aims to improve transmission cost efficiency and the reliability of remote health monitoring systems [18,19].

The major contributions of this study are summarised as follows:

- We provide an extensive evaluation of adopted commonly used DL models in computer vision (such as Resnet, VGG, Alexnet) and time series analysis (such as BILSTM, GRU, LSTM) [20,21] to develop an ECG noise detection and classification model. To align the 1d nature of ECG signals, we convert the adopted 2d CNN-based DL to a 1d CNN and fine-tune (such as batch size, learning rate and number of epochs) DL models to maximise the ECG noise detection and classification method. To the best of our knowledge, this is the first attempt to compare all popular DL models in computer vision and time series analysis for ECG noise detection and classification in terms of model complexity and performance.
- The performance of these models is evaluated using six dataset sources from Physionet, including BIDMC, CINC-2011, CINC-2014, ECG-ID, MIT arrhythmia, and teleECG to assess the generalisation capability of these DL models.
- We propose a practical IoMT-based framework to showcase the usability of ECG noise detection and classification, which is missing in existing studies. Implementing DL-based ECG noise detection and classification in an IoMT framework is essential to ensure compatibility with computation resources.
- Our experimental results show that the adopted popular DL model architecture, such as Resnet and VGG, is superior in ECG classification accuracy (97.31% and 95.07%) and enhances the robustness of noise detection and classification techniques in clinical and non-clinical settings. These robust systems contribute to developing a knowledge-based filtering system, effectively filtering only noisy ECG signals.

The rest of the paper is organised as follows. In Section 2, we describe the system design, methodology, used models, and experimental setup. Section 3 presents experimental results in detail with diverse datasets. The findings from the result section are summarised in Section 4. Finally, in Section 5, we conclude the paper and discuss future works.

1.1. An end-to-end smart ECG noise classification framework

In this section, we propose an end-to-end smart ECG noise classification framework in the context of an IoMT system. Then, we provide a formal description of the proposed framework.

• **Framework Overview:** Let us consider an end-to-end situation where the ECG signal of a patient is transferred to an automatic ECG analyser for a doctor's analysis (as shown in Fig. 2).

The signals can be generated from inhouse, dynamic and remote healthcare settings. Signals are collected from sensor devices (such as wired and wireless Internet of Things (IoT) devices). These signals are transferred through a smart ECG filtering system that is deployed in the IoT-enabled gateways. This paper proposes a novel end-to-end knowledge-based ECG noise filtering framework using a DL-based ECG noise classification approach. Note that, in our architecture, the employed DL-model resides within the IoT gateway. The filtered signals can be transmitted through intra (i.e., within a close inhouse setup) or Internet (i.e., in remote setup) gateway for storage. The stored signals are analysed, and appropriate visualisation is performed to generate patient reports. We use internet gateways to transmit ECG signals to doctors in different locations. Significantly, in this paper, we consider the situation where the ECG signals are captured in a 'signal filtering system' deployed in IoT gateways before they are sent to the remote ECG analysers (a simple process of such transmission is shown in Fig. 3).

The effectiveness of the proposed framework lies in the fact that the framework saves the transmission cost and storage (such as in cloud, or inhouse) by transmitting only clean ECG signals.

• **A Formal Description of the Proposed Framework:** Our proposed framework combines a knowledge-based filtering system using ECG noise detection and classification in intra and internet gateways. The novel part of the proposed framework is highlighted with a dotted line in Fig. 2. Let us assume the ECG signal, $X_{clean} = \{x_1, x_2, \ldots, x_n\}$, coming

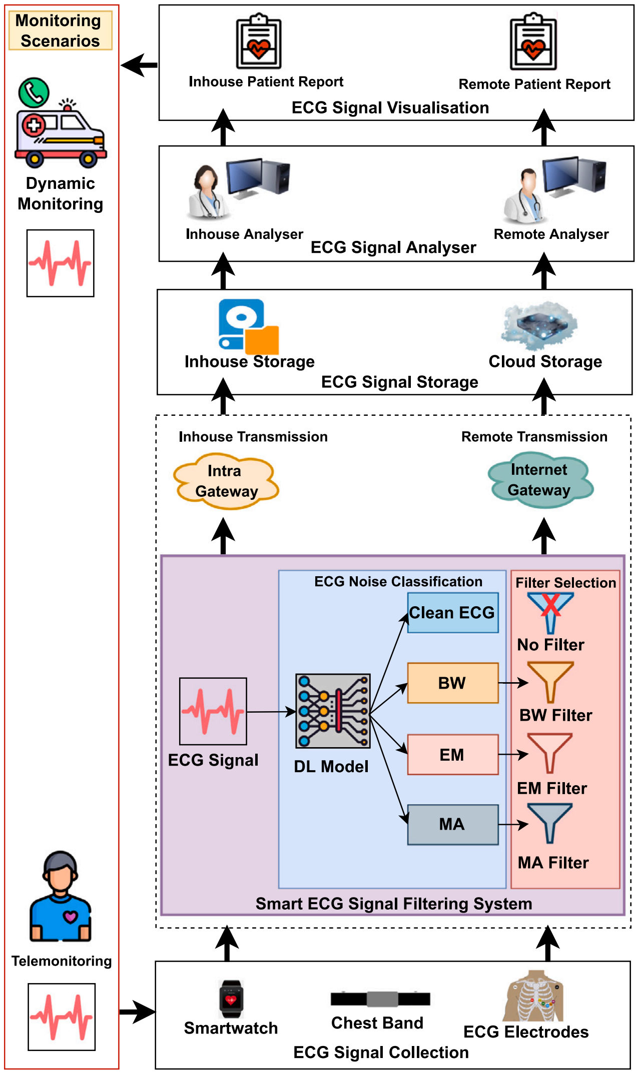

Fig. 2. A typical end-to-end setup from collection and visualisation of ECG signals using traditional inhouse, dynamic, and remote monitoring scenarios. The highlighted dotted line denotes the key contributions of our paper.

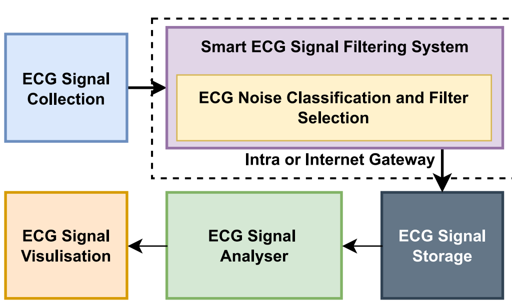

Fig. 3. A simple end-to-end view of the process from collecting ECG signals to visualisation as shown in Fig. 2. Our contribution domain is highlighted in the dotted line.

from ECG devices, where n denotes the length of the ECG. During ECG collection, the ECG signal, X_{clean}, is contaminated with different types of noise, such as, muscle artefacts (MA), baseline wander (BW), and electrode motion (EM), denoted as noisy ECG, $X_{noise} = X_{clean} + MA$ or $X_{noise} = X_{clean} + EM$ or $X_{noise} = X_{clean} + BW$. Now, we want to remove this noise from the ECG signal using a knowledge-based filter, K_{filter}, utilising ECG noise detection and classification methods using different DL models, $M_{DL} = \{m_{VGG}, m_{Resnet}, m_{Alexnet}, m_{LSTM}, m_{GRU}, m_{BILSTM}\}$. After fine-tuning all the adopted DL models using a set of publicly available ECG datasets, $\mathcal{D} = \{d_1, d_2, \dots, d_j\}$, where $j = 6$ datasets, we obtain the optimum model, O_{DL}, for the K_{filter} method. The K_{filter} method can be expressed as follows:

$$K_{filter} = f(X_{clean} \text{ or } X_{noise}, O_{DL}) \tag{1}$$

where $f(X_{clean}$ or $X_{noise}, O_{DL})$ is a function for predicting types of noise. Once the noise type is predicted, a specific noise removal filter will be used to clean the ECG signals instead of blind filtration (applying filters without knowing the ECG signal condition) or being removed from further analysis. This K_{filter} (1) method will reduce computational and transmission costs by using a specific filter and removing extremely noisy ECG signals for further analysis.

1.2. Related work

In this section, we briefly review the existing studies (along with their limitations) in three categories, namely, (i) threshold, (ii) ML, and (iii) DL-based ECG noise detection and classification, as indicated in Section 1.

• **(i) Threshold-Based ECG Noise Detection and Classification:** Several works show the early ECG noise detection that was done using statistical features with thresholds. For example, *Satija et al.* [22] proposed an ECG noise detection system based on discrete Fourier transform (DFT) using set specific threshold values and reported sensitivity 95.56% and specificity 97.85% for two publicly available and one private dataset including simulated noise. *Zhidong et al.* [23] introduced a heuristic fusion and fuzzy-based model where ECG segments were treated as a noisy ECG when crossing the specific amplitude and achieved accuracy 94.67% for two publicly available datasets. In 2019, authors adopted variable frequency complex demodulation (VFCDM) based on time–frequency spectral (TFS), where the noisy signal is detected using a range of threshold values which reduces 94% false positive for 60 subjects from MIMIC III dataset [24]. In addition, *Satija et al.* [7] proposed a modified ensemble empirical mode decomposition (CEEMD) model for automatic noise detection from ECG signals and reported sensitivity 99.12% and specificity 98.56% for simulated noisy datasets. However, appropriate threshold selection and the small scale and simulated dataset limited the clinical applicability shown in [6].

• **(ii) ML-Based ECG Noise Detection and Classification:** The threshold-based ECG noise detection limitation can be mitigated by using the classical ML models from features. For example, *Tobon et al.* [25] utilised the modulation spectral quality index (MS-QI) feature to train SVM and LDA Ml models and achieved accuracy 95% and 96%, respectively, for simulated and two publicly available datasets. However, they did not mention the performance of the third dataset in their study. *Li et al.* [26] proposed an SVM-based ECG noise detection system and found 57.26% accuracy for the MIT arrhythmia dataset. Another group of authors proposed an SVM model for Lenovo h3 devices and got 96.1% accuracy. However, the model is not validated using diverse datasets. In [8], *Liu el al.* designed an IoT-based ECG noise detection system and achieved sensitivity 97.9% and specificity 96.4%, respectively. However, their model included cloud computing, where offline noise detection is impossible. In [27], *Abbasi et al.* proposed a multiclass SVM model for noise detection using simulated noisy ECG datasets and achieved 86.61% accuracy.

These ML models need handcrafted features from raw ECG signals before training the model. So, the performance of the ML model depends on feature selection, and wrong feature selection can increase the false positive or negative rates. Real-time ECG noise detection in an edge device (such as a smartphone) should be a resource constraint and energy efficiency. However, the ML model requires two steps of processing to prepare input data: data collection and feature extractions, which increases the computational cost.

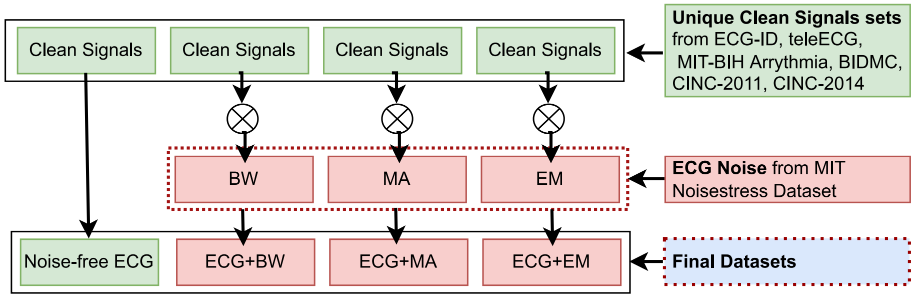

Fig. 4. The preparation of datasets for ECG noise detection and classification involved several key steps. Initially, we carefully segmented four sets of clean ECG (green colour boxes) signals derived from six distinct datasets: ECG-ID, teleECG, BIDMC, MIT-BIH arrhythmia, CINC-2011, and CINC-2014. Ensuring there was no data leakage between these sets. Subsequently, we combine various types of ECG noise (red colour boxes) with clean ECG signals to create noisy ECG signals for final datasets (blue colour box) to train and testing the DL model. This step was significant in simulating real-world scenarios and enhancing the model's ability to perform well on unseen ECG signals.

• **(iii) DL-Based ECG Noise Detection and Classification:** DL models are widely used in computer vision for promising results and extraction of abstract features without manual intervention. Currently, researchers are exploring a DL model in time-series datasets, such as PPG, ECG and EEG. There are several papers on ECG noise detection using a DL model. For example, *Zhang et al.* [28] proposed a Cascaded CNN-based noise detection system combining ECG images and signals using private and one publicly available datasets and reported 91.80% accuracy. *Kido et al.* [11] introduced a convolutional neural network (CNN)-based qua_{model} for ECG noise detection for only 12 subjects, and his model needs to validate for diverse datasets. In 2022, *Jin et al.* [13] proposed a DAC-LSTM ECG noise detection system and achieved 95% accuracy for one dataset only.

In summary, the studies mentioned above lack validating models for diverse datasets and hence, the robustness of these models is questionable. In our study, we employ DL models based on RNN and CNN for detecting and classifying noisy ECG signals. Performance of these models were tested using a diverse dataset to evaluate robustness and we made a comparison between CNN and RNN based models.

2. Methods

In this section, we discuss (i) the data sets used for the experiments, (ii) the methodology, (iii) the models used, and (iv) the experimental setup.

2.1. Datasets

In this study, six publicly available datasets, including ECG-ID, BIDMC, CINC-2011, CINC-2014, teleECG and MIT-BIH arrhythmia [29], are used in this study to validate the ECG noise detection system. These datasets were manually annotated based on the accepted definitions of noisy and noise-free ECG [8]. According to that definition, an ECG segment is considered noisy when the ECG signal is contaminated by any disturbances, including muscle artefacts (MA), baseline wander (BW), and electrode motion(EM). Additionally, if the QRS complex and T wave are clearly discernible, an ECG segment is noise-free. In our study, to prevent data leakage and ensure robust model evaluation, we meticulously divided each dataset's clean ECG signals into four non-overlapping groups. This strategy guarantees that the model is assessed on unseen data during testing. Furthermore, when generating noisy ECG signals by combining clean ECG segments with MIT Noisestress [30] noise types, we ensure that these noise types are not superimposed on each other, as illustrated in Fig. 4. This meticulous approach safeguards against the introduction of artificial correlations that could potentially mislead the model. In this study, we created a semi-automatic technique shown in Fig. 4 for labelling every ECG segment and summarised it in Table 1. In the DL model, the length of the segments should be constant across the datasets. To provide a constant signal length for the DL model, all annotated datasets were transformed to 500 Hz sampling frequency. We also scale all the ECG segments to reduce the amplitude variation of ECG segments for the diverse datasets using min–max normalisation and can be expressed as:

Table 1
Summary of the number of 10-s raw noise-free ECG segments and total recording length for ECG-ID, BIDMC, CINC-2011, CINC-2014, TeleECG, and MIT-BIH arrhythmia datasets. Additionally, the table includes the count of individual noise-type segments and their respective total recording lengths.

Dataset	No. of segments	Total recording length (Hour)
ECG-ID	128	0.36
BIDMC	2372	6.59
CINC-2011	1022	2.84
CINC-2014	4871	13.53
TeleECG	231	0.64
MIT-BIH arrhythmia	8503	23.62
Total number of 10s segments for individual types of signal		
Noise-free ECG	4208	11.69
ECG+BW	4240	11.78
ECG+EM	4293	11.93
ECG+MA	4386	12.18
Total (4Ns)	17 127	47.58

$$Y = \frac{X - min(X)}{max(X) - min(X)} \tag{2}$$

where, $X = \{x_1, x_2, \ldots, x_n\}$ is the ECG segment, n is the length of X, and Y is the normalised segment values between 0 and 1.

Algorithm 1: DL-based ECG noise detection and classification.

Input: $X(n) \rightarrow$ 10s ECG segment
$Model_{DL} \rightarrow$ [Resnet, Alexnet, VGG, LSTM, GRU, and BILSTM]
Output: $D \rightarrow$ (Best model)
Step 1: Min-max scaling of 10s ECG segment ($X(n)$)
for $k \in \{1, \ldots, n\}$ **do**
  Scale ECG, $X[k] = \frac{X[k]-min(X)}{max(X)-min(X)}$
Step 2: Train and test DL with six ECG datasets
for $i \in Model_{DL}$ **do**
  for $j \in leaveOneout(Sixdatasets)$ **do**
    $i.fit(fivedatasets)$ $predict_{class}, \gamma = i.predict(onedatasets)$
    $Acc, \beta = Accuracy(true_{class}, \gamma)$
    $Model_{complexity}, \lambda = i.countParameter()$
Step 3: Optimum model, $O_{DL} = max(\beta)$ and $min(\lambda)$

2.2. Experimental protocol

The employed ECG noise classification algorithm (i.e., Algorithm 1) involves a series of steps as follows:

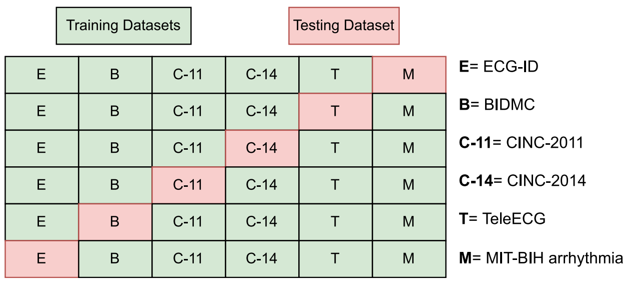

Fig. 5. Overview of leave one training (green colour boxes) and testing (red colour box) datasets with 10% validation data from training datasets.

- **Input:** All six ECG dataset signals are normalised to the range from 0 to 1. After resampling all the ECG signals at 500 Hz to make all the ECG signal the same length as a requirement of the DL model.
- **Output:** Best DL model for ECG noise classification by making a tradeoff between $Model_{complexity}(\lambda)$ and $Acc(\beta)$.
- **Train and test DL model:** Six DL models from RNN and CNN categorise are selected to train and test the model using leave one dataset out to ensure no data leakage. To ensure robust evaluation and generalisability, we avoid data leakage by not utilising the same dataset for both training and testing our proposed model. This approach guarantees that the model is assessed on unseen data, fostering its ability to perform well on datasets beyond the one used for training.
- **Compare models' performance and complexity:** Compare six models' performance and model complexity in terms of the number of parameters and generalisation capabilities.

2.3. Models

In this study, we consider two types of DL models, one is (Recurrent Neural Network) RNN-based, and another is (Convolutional Neural Network) CNN-based.

- **RNN-Based DL Model:** LSTM, GRU, and BILSTM are types of RNN designed to handle the problem of vanishing gradients in traditional RNNs. LSTM, GRU, and BILSTM architecture consist of several gates controlling the information flow into and out of the memory cell, including an input gate, an output gate, and a forget gate. These gates are trained to learn which information is essential to keep and which should be discarded based on the memory cell's current input and previous state. These models are famous for time-series [21,31].
- **CNN-Based DL Model:** CNN architectures, such as Alexnet, VGG, and Resnet, are widely used for image classification and related applications. Six convolutional layers and three fully linked layers make up Alexnet, and it proposed activation functions for ReLU and local response normalisation (LRN) layers to boost performance. Small 3×3 convolutional filters are used in each layer of the VGG's up to 19 layers to extract features from the input. Resnet, which stands for 'Residual Network', is an extremely complex CNN design that introduced the idea of residual connections, which let the network skip some layers while keeping data from earlier layers. With hundreds of layers in very deep architectures, this makes it simpler for the network to learn meaningful representations [20,32].

All six architectures mentioned above have been used extensively for various image classification and related tasks in computer vision. We have been motivated to use these models to validate their contribution in time series instead of computer vision and how they adapt to new data structures and complexity.

2.4. Experimental setup

Recall, in this study, we use the six most popular CNN-based architectures in computer vision called VGG, Alexnet, and Resnet by converting architectures into 1d CNN format and time-series analysis DL model architecture, such as GRU, LSTM and BILSTM shown in Fig. 6. All the models were trained on six different ECG datasets (ECG-ID, BIDMC, CINC-2011, CINC-2014, teleECG and MIT-BIH arrhythmia). We test the model's performance using the leave one dataset out technique to ensure there is no data leakage between training and testing datasets, as shown in Fig. 5. Fig. 3 depicts our approach in ECG noise classification as a pipeline.

Employing a server to conduct the training process is advantageous since building an RNN, and CNN model demands substantial computer resources. Here is a list of setups that we follow in our experiments: (i) *Hardware:* we use a Linux server with a Tesla V100 GPU, which has 32.51 GB of memory, (ii) *Software:* we use python, and the RNN and CNN model is created using a Tensorflow environment, finally, (iii) *Data Preparation:* before training the CNN model, the data must be preprocessed and organised in a format suitable for training. We have resampled the ECG signal into 500 Hz as the original datasets are not in the same sampling frequency.

3. Results

In this section, we discuss the achieved results. We show the (i) individual class performance, (ii) overall class performance, and (iii) model complexity. They are as follows:

3.1. Individual class performance across the datasets

Significantly, we have used six datasets with four classes in the experiment. They are as follows:

- LSTM: ECG-ID: Clean, BW, EM and MA signal type achieved 68.75%, 75.78%, 60.94% and 66.41% accuracy, whereas F1 score is 51.22%, 0.0%, 37.50% and 31.75%.
- BILSTM: CINC-2011: Clean, BW, EM and MA signal type achieved 75.54%, 69.28%, 69.08% and 70.06% accuracy, whereas F1 score is 56.60%, 38.19%, 18.56% and 46.50%.
- GRU: ECG-ID: Clean, BW, EM and MA signal type achieved 71.88%, 75.78%, 52.34% and 73.44% accuracy, whereas F1 score is 51.35%, 0.0%, 47.86% and 0.0%.
- Alexnet: ECG-ID: Clean, BW, EM and MA signal type achieved 98.44%, 99.22%, 100.0% and 97.66% accuracy, whereas F1 score is 96.77%, 98.36%, 100.0% and 95.65%.
- VGG: ECG-ID: Clean, BW, EM and MA signal type achieved 100.0%, 99.22%, 100.0% and 99.22% accuracy, whereas F1 score is 100.0%, 98.36%, 100.0% and 98.55%.
- Resnet: CINC-2014: Clean, BW, EM and MA signal type achieved 99.94%, 96.3%, 96.57% and 99.75% accuracy, whereas F1 score is 99.87%, 92.92%, 92.79% and 99.52%.

Performance of DL model for individual class and dataset shown in Fig. 7. We have also highlighted the performance level using bold text. More than 90% performance level has been highlighted with bold text in Table 2. From Table 2, it is evident that CNN-based models showed mostly 90% performance level than RNN-based models.

3.2. Overall class performance

We conducted an experiment to investigate the performance and model complexity of six well-known CNN and LSTM-based models for ECG noise detection and classification. From Fig. 8, LSTM model demonstrated an average accuracy of 65.90% and an F1 score of 25.37%, while the GRU model achieved an accuracy of 65.76% and

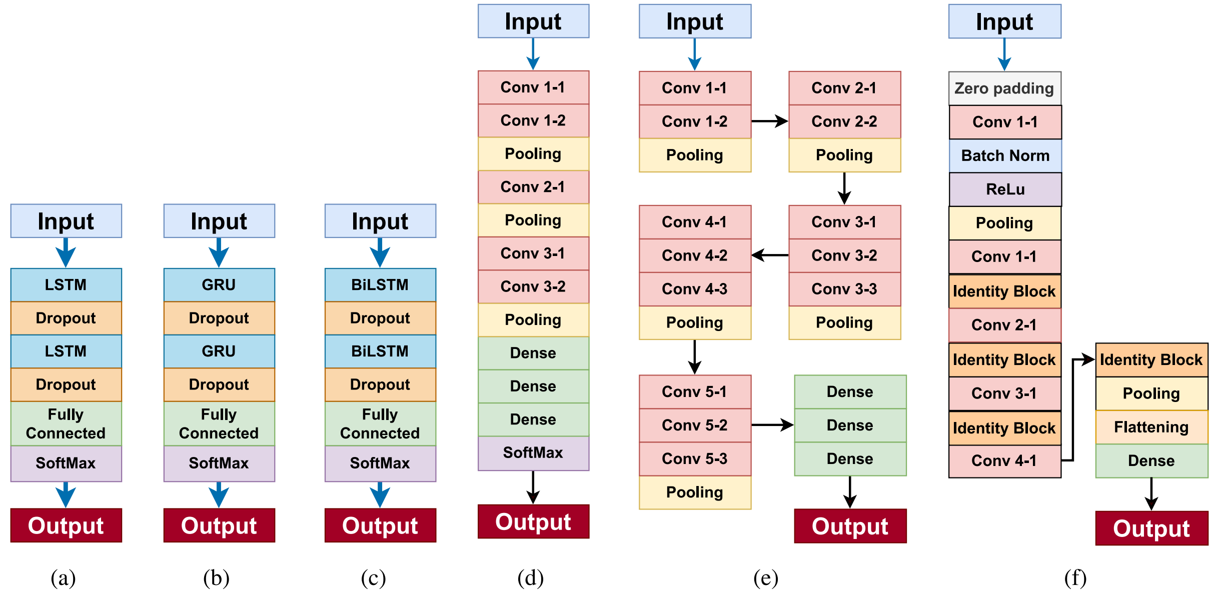

Fig. 6. The overview of adopted popular RNN and CNN-based DL architecture used in this study: (a) LSTM, (b) GRU, (c) BILSTM, (d) Alexnet, (e) VGG, and (f) Resnet.

Table 2
We employed a comprehensive evaluation strategy to assess the performance of various models. This strategy involved comparing the accuracy and F1 scores (%) across diverse datasets and classes. Models achieving higher scores are denoted by the **bold** text.

Model	Datasets	Signal types							
		Clean		Baseline Wander (BW)		Electrode Motion (EM)		Muscle Artefacts (MA)	
		Accuracy (%)	F1 (%)	Accuracy (%)	F1 (%)	Accuracy (%)	F1 (%)	Accuracy (%)	F1 (%)
LSTM	BIDMC	50.89	46.19	73.78	15.26	72.47	19.28	69.81	30.89
	CINC-2011	55.77	48.17	73.19	14.91	73.87	9.49	66.63	38.56
	CINC-2014	63.29	36.23	75.69	0.00	45.70	41.24	74.28	0.00
	ECG-ID	68.75	51.22	75.78	0.00	60.94	37.50	66.41	31.75
	MIT arrhythmia	55.63	45.10	66.58	17.58	62.73	30.37	74.90	0.00
	teleECG	64.50	39.71	55.41	16.26	73.16	3.12	61.47	35.97
GRU	BIDMC	62.86	48.45	73.27	9.43	56.96	39.62	72.85	0.00
	CINC-2011	59.78	46.69	69.86	10.98	56.36	33.43	74.85	0.00
	CINC-2014	62.12	38.56	75.69	0.00	48.84	41.09	73.25	1.66
	ECG-ID	71.88	51.35	75.78	0.00	52.34	47.86	73.44	0.00
	MIT arrhythmia	58.04	45.09	74.82	0.00	53.30	36.29	74.90	0.00
	teleECG	58.44	42.17	77.06	0.00	49.35	34.64	72.29	0.00
ALEXNET	BIDMC	**98.19**	**96.35**	**93.25**	**84.03**	**93.38**	**87.80**	**98.15**	**96.55**
	CINC-2011	**98.43**	**96.79**	**97.85**	**95.65**	**98.24**	**96.48**	**97.46**	**95.08**
	CINC-2014	82.71	45.40	95.13	89.05	98.25	96.52	77.34	69.28
	ECG-ID	**98.44**	**96.77**	**99.22**	**98.36**	**100.00**	**100.00**	**97.66**	**95.65**
	MIT arrhythmia	76.18	8.33	89.92	81.76	90.50	78.77	70.36	59.95
	teleECG	77.06	18.46	87.01	75.81	94.81	88.24	77.06	69.01
VGG	BIDMC	**99.58**	**99.13**	**91.19**	**77.88**	**89.80**	**81.78**	**98.69**	**97.65**
	CINC-2011	**99.90**	**99.80**	**98.04**	**96.21**	**98.43**	**96.79**	**99.12**	**98.23**
	CINC-2014	**99.98**	**99.96**	**98.91**	**97.80**	**98.91**	**97.82**	**99.77**	**99.56**
	ECG-ID	**100.00**	**100.00**	**99.22**	**98.36**	**100.00**	**100.00**	**99.22**	**98.55**
	MIT arrhythmia	**99.71**	**99.41**	**99.07**	**98.14**	**99.01**	**98.03**	**99.81**	**99.63**
	teleECG	**97.40**	**94.64**	**83.98**	**54.32**	**85.71**	**76.60**	**100.00**	**100.00**
RESNET	BIDMC	96.59	92.44	87.14	69.83	87.39	77.84	98.78	97.79
	CINC-2011	**99.90**	**99.80**	**96.18**	**92.51**	**96.28**	**92.58**	**98.83**	**97.61**
	CINC-2014	**99.94**	**99.87**	**96.30**	**92.92**	**96.57**	**92.79**	**99.75**	**99.52**
	ECG-ID	100.00	100.00	95.31	89.29	91.41	81.36	91.41	86.08
	MIT arrhythmia	99.99	99.98	97.92	95.99	97.89	95.67	99.89	99.79
	teleECG	93.07	84.31	80.95	42.11	80.09	70.51	100.00	100.00
BILSTM	BIDMC	66.91	51.33	72.89	22.25	58.60	38.47	73.02	9.60
	CINC-2011	75.54	56.60	69.28	38.19	69.08	18.56	70.06	46.50
	CINC-2014	75.30	23.52	75.53	21.48	62.18	43.77	55.96	36.44
	ECG-ID	71.09	51.95	75.00	20.00	67.19	32.26	61.72	36.36
	MIT arrhythmia	64.37	49.52	68.40	33.51	60.78	30.13	74.88	2.47
	teleECG	66.67	37.40	57.14	18.18	69.70	7.89	61.90	38.03

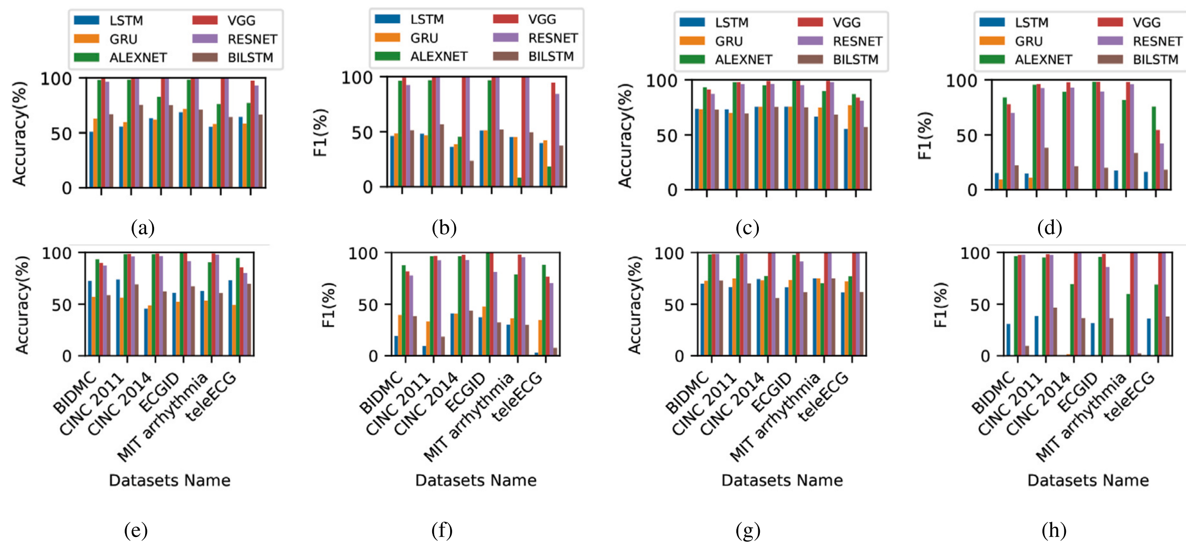

Fig. 7. The performance of different DL models using accuracy and F1 performance metrics. (a) to (b): clean signal; (c) to (d): baseline wander (BW); (e) to (f): electrode motion (EM); (g) to (h) motion artefact (MA). Across all class types, VGG, Alexnet and Resnet outperform compared to other models, especially the VGG model. Across the different classes and datasets, VGG and Resnet models outperform other DL models.

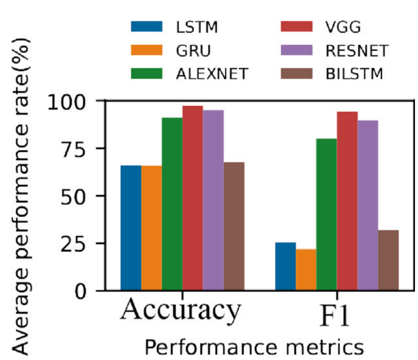

Fig. 8. Overall accuracy and F1 score of all DL models. CNN-based models (VGG, Alexnet and Resnet) perform better (more than 70%), especially the VGG model, than other models (GRU, LSTM and BILSTM).

Table 3
Comparison of DL model training time and a number of parameters.

Model name	Model parameter	Training time [Hours]
LSTM	147 364	5.98
GRU	111 284	5.56
ALEXNET	179 477 772	0.36
VGG	181 154 628	0.19
RESNET	523 592	0.24
BILSTM	38 692	2.53

an F1 score of 21.97%. Among all the RNN models, the BILSTM model had the highest average accuracy and F1 score, at 67.63% and 31.85%, respectively. From Fig. 8, Alexnet model achieved an average accuracy and F1 score of 91.11% and 80%, respectively. The VGG model achieved an average accuracy and F1 score of 97.31% and 94.18%, respectively, while the Resnet model achieved an average accuracy and F1 score of 95.07% and 89.61%, respectively. The CNN-based DL models outperformed the RNN-based models.

3.3. Model complexity

LSTM, GRU, and BILSTM model parameters are 147 364, 111 284 and 38 692 respectively, whereas training time is 5.98, 5.56, and 2.53 h as shown in Table 3. The high training time is because of their architecture, which involves processing the input data across multiple timesteps, and the network has to learn the appropriate weights for each of these timesteps, which can be computationally intensive, especially if the sequence length is relatively long. In our case sequence length is 5000 (equal to 10 s ECG segments).

For Alexnet, VGG and Resnet model parameters are 179 477 772, 181 154 628 and 523 592, respectively, where training time is 0.36, 0.19 and 2.53 h respectively, as shown in Table 3. Resnet shows higher training time because of the number of layers, and each layer has a convolution operation, which makes the model computationally expensive.

4. Discussion

Automatic ECG signal analysis depends on ECG data quality. Poor quality of signal creates false alarms and misguides the automated ECG analyser. In addition, the noisy ECG increases transmission costs by sending poor-quality ECG signals. To address poor-quality ECG signals, blind filtering approaches are used in most of the pre-processing steps in an ECG analyser. However, inappropriate filter frequency and order distort the ECG characteristics. Nevertheless, the knowledge base filter can reduce this distortion by classifying noise types before applying a specific filter. DL model can be used to develop a knowledge-based filter. We consider DL instead of the ML model due to dependence on handcrafted features. In this study, we evaluated different DL (RNN and CNN-based models) and their complexity to save storage and computational cost. The paper presents a knowledge-based ECG filtering system employing DL models for ECG noise classification and comparing computer vision models in an IoMT framework.

4.1. Individual class performance across the datasets

By evaluating the performance of individual classes, it is easy to know which classes are accurately classified using a specific model or technique. This evaluation also helps determine which classes are more challenging to classify. From Figs. 7(a) and 7(b), clean ECG signal performs better for CNN-based models, such as Alexnet, Resnet and VGG. However, Alexnet performs poorly for CINC-2014, teleECG and MIT arrhythmia datasets. For BW noise classification Alexnet perform well for teleECG compared to other models shown in Figs. 7(c) and 7(d). For EM noise classification Alexnet and VGG perform better than Resnet, illustrated in Figs. 7(e) and 7(f). On the other hand, VGG and Resnet perform better for MA noise classification compared to Alexnet, shown in Figs. 7(g) and 7(h).

Table 4
Comparison of existing study and our proposed model for ECG noise detection and classification. Compared to existing studies, we have used diverse datasets to generalise the DL model. Previous studies mostly focus on ECG noise detection. Unlike them, in this study, we work together on ECG noise detection and classification.

Ref. (Models)	Year	Datasets	Noise detection	Noise classification	Accuracy %
[12]	2022	CINC-2011	✓	✗	94.55
[14]	2022	MIT arrhythmia	✓	✗	100.00
[33]	2023	CINC-2011	✓	✗	95.00
[34]	2021	CINC-2011	✓	✗	93.09
[35]	2018	CINC-2011	✓	✓	96.45
[Our work]	**2024**	BIDMC	✓	✓	99.58
		CINC-2011	✓	✓	99.90
		CINC-2014	✓	✓	99.98
		ECG-ID	✓	✓	100.00
		MIT arrhythmia	✓	✓	99.99
		teleECG	✓	✓	97.40

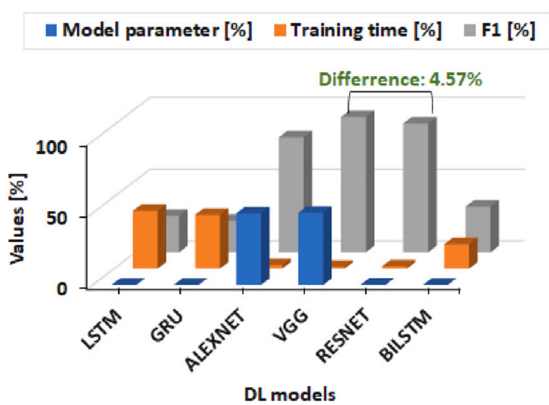

Fig. 9. Relative training time, model parameters and F1 score. LSTM, GRU, and BILSTM models are lightweight but have low F1 scores compared to CNN-based models. Alexnet and VGG model complexity are similar, but the F1 score is high for VGG. Compared to VGG and Resnet, the F1 score is 4.57% higher in VGG, but model complexity is 49.98% higher than Resnet.

4.2. Overall class performance

The individual result shows accuracy and F1 score fluctuation regarding six datasets. To see the overall accuracy and F1 score, we averaged the individual class performance for CNN and RNN-based DL models. Fig. 8 shows that CNN-based model (above 75%), especially the VGG model, performs better than other models. The overall performance comparison of existing studies and our work is shown in Table 4.

4.3. Model complexity

The DL model is better than ECG noise classification, but its complexity restricts deployment on low-power devices. As shown in Fig. 9, VGG and Resnet perform better than other models, but VGG has a higher complexity than Resnet. Our experiments show that Resnet is suitable for resource-constrained environments by considering a 4.57% F1 score less than VGG.

Our study demonstrates the complexity of the CNN and RNN-based ECG noise classification models and their performance. This finding will help decide which DL model fits ECG noise classification and which is less complex. Less complex model speed up classification response and help take quick decisions about ECG abnormalities.

5. Conclusions

In this paper, we have proposed a DL-based ECG noise classifier to create a knowledge-based ECG filtering system. Unlike blind filtering ECG signals, the ECG noise classification outcome can be used for the selection and application of specific filters based on the presence or types of ECG noise. This type of selective filtering approach will reduce the ECG signal distortion due to filter order and cut-off frequency. We have extensively evaluated the proposed model with diverse datasets and different CNN and RNN-based DL models based on performance and model complexity. We present a DL-based knowledge-driven ECG filtering system that classifies ECG noise types and evaluates computer vision model architectures within a practical IoMT framework. Our results showed that VGG and Resnet perform well compared to other models. VGG model is superior in terms of performance. But Resnet performs comparably to VGG with less model complexity. This gives users a choice on the DL model according to their available computational power and device memory. In future, we plan to deploy the DL models used in this paper in smartphones and micro-controllers to examine their performance comparison. In addition, combining different types of noise will be beneficial for a more robust ECG noise classification model in complex scenarios. However, we leave this for another future research work.

CRediT authorship contribution statement

Saifur Rahman: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shantanu Pal:** Writing – review & editing, Supervision, Methodology. **John Yearwood:** Writing – review & editing, Supervision, Methodology. **Chandan Karmakar:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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